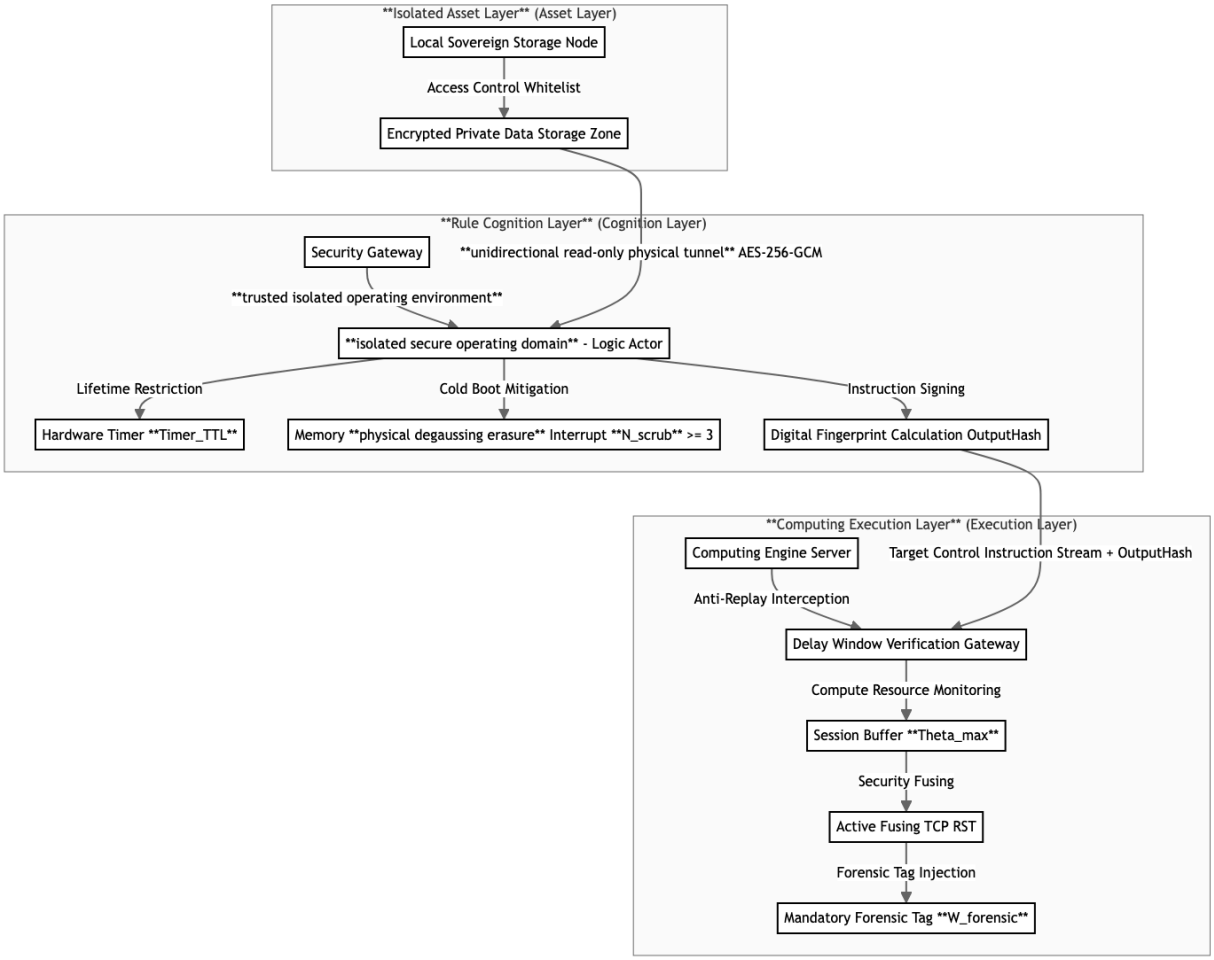

Isolated Asset Layer (Asset Layer)
Local Sovereign Storage Node
Access Control Whitelist
Encrypted Private Data Storage Zone
Rule Cognition Layer (Cognition Layer)
Security Gateway
unidirectional read-only physical tunnel AES-256-GCM
trusted isolated operating environment
isolated secure operating domain - Logic Actor
Lifetime Restriction
Cold Boot Mitigation
Instruction Signing
Hardware Timer **Timer_TTL**
Memory **physical degaussing erasure** Interrupt **N_scrub** >= 3
Digital Fingerprint Calculation OutputHash
Computing Execution Layer (Execution Layer)
Computing Engine Server
Target Control Instruction Stream + OutputHash
Anti-Replay Interception
Delay Window Verification Gateway
Compute Resource Monitoring
Session Buffer **Theta_max**
Security Fusing
Active Fusing TCP RST
Forensic Tag Injection
Mandatory Forensic Tag **W_forensic**

图1

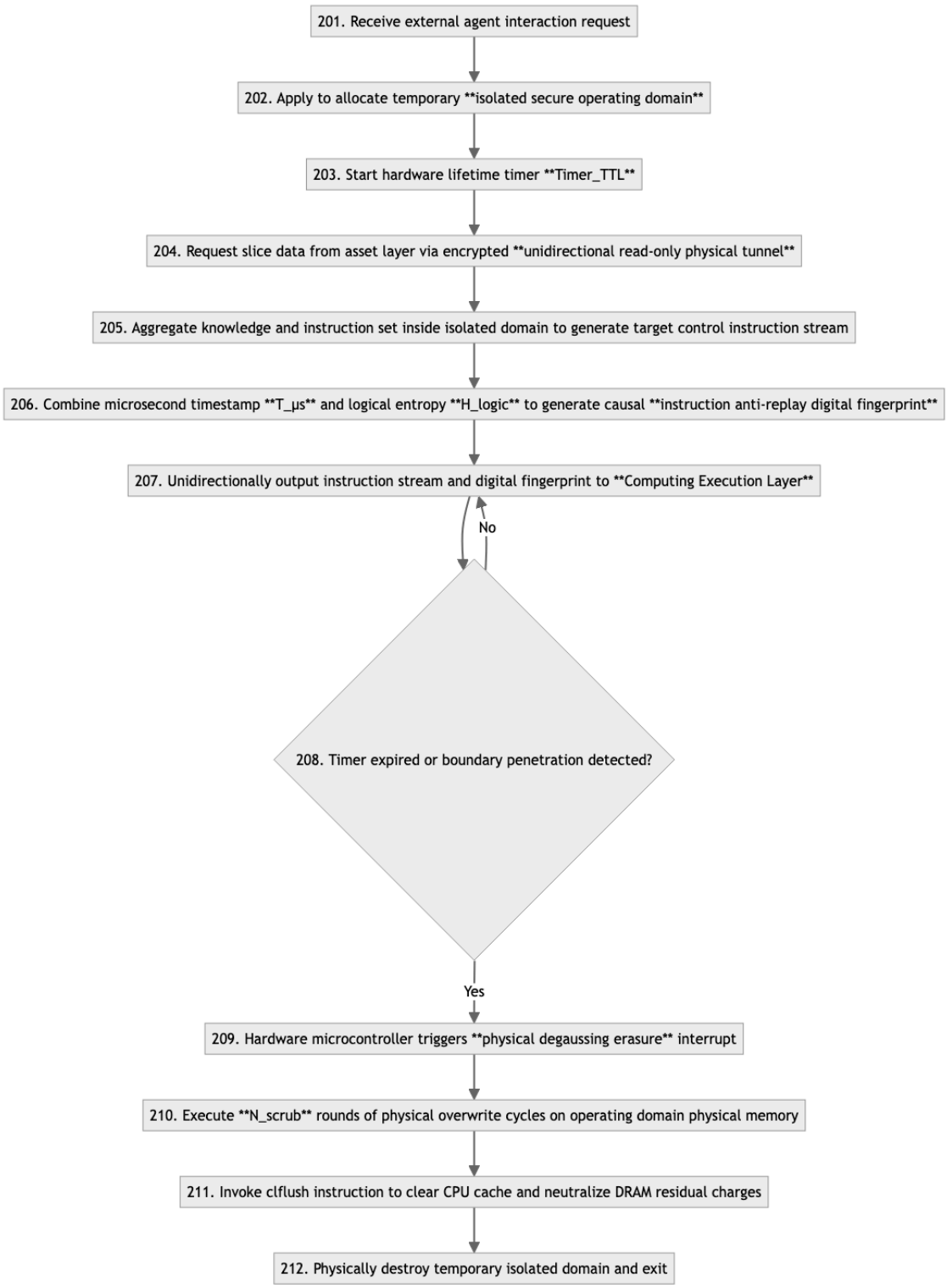

201. Receive external agent interaction request
202. Apply to allocate temporary **isolated secure operating domain**
203. Start hardware lifetime timer **Timer_TTL**
204. Request slice data from asset layer via encrypted **unidirectional read-only physical tunnel**
205. Aggregate knowledge and instruction set inside isolated domain to generate target control instruction stream
206. Combine microsecond timestamp **T_μs** and logical entropy **H_logic** to generate causal **instruction anti-replay digital fingerprint**
207. Unidirectionally output instruction stream and digital fingerprint to **Computing Execution Layer**
208. Timer expired or boundary penetration detected?
No
Yes
209. Hardware microcontroller triggers **physical degaussing erasure** interrupt
210. Execute **N_scrub** rounds of physical overwrite cycles on operating domain physical memory
211. Invoke clflush instruction to clear CPU cache and neutralize DRAM residual charges
212. Physically destroy temporary isolated domain and exit

图2



图3

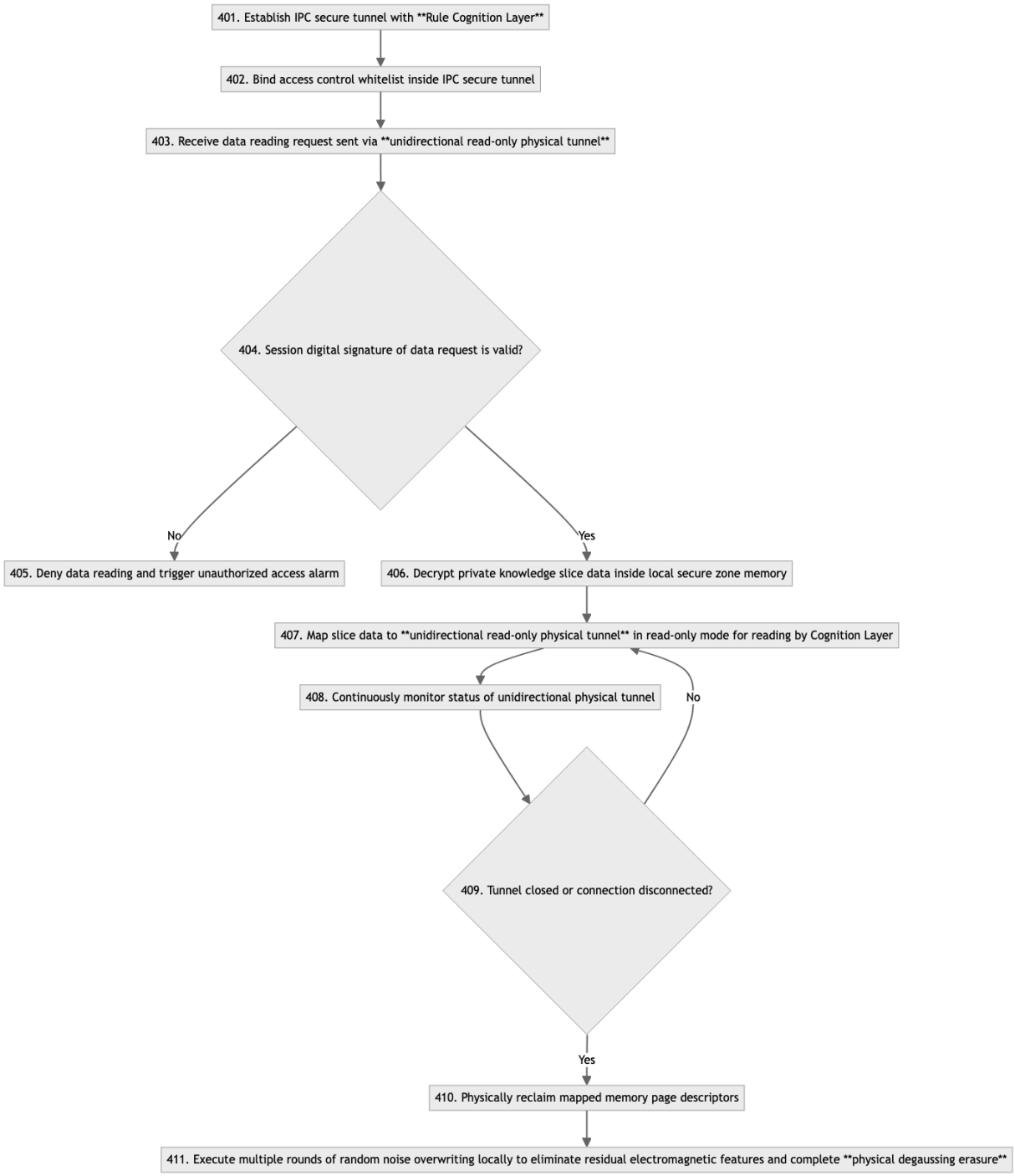

401. Establish IPC secure tunnel with **Rule Cognition Layer**
402. Bind access control whitelist inside IPC secure tunnel
403. Receive data reading request sent via **unidirectional read-only physical tunnel**
404. Session digital signature of data request is valid?
No
Yes
405. Deny data reading and trigger unauthorized access alarm
406. Decrypt private knowledge slice data inside local secure zone memory
407. Map slice data to **unidirectional read-only physical tunnel** in read-only mode for reading by Cognition Layer
408. Continuously monitor status of unidirectional physical tunnel
No
409. Tunnel closed or connection disconnected?
Yes
410. Physically reclaim mapped memory page descriptors
411. Execute multiple rounds of random noise overwriting locally to eliminate residual electromagnetic features and complete **physical degaussing erasure**

图4